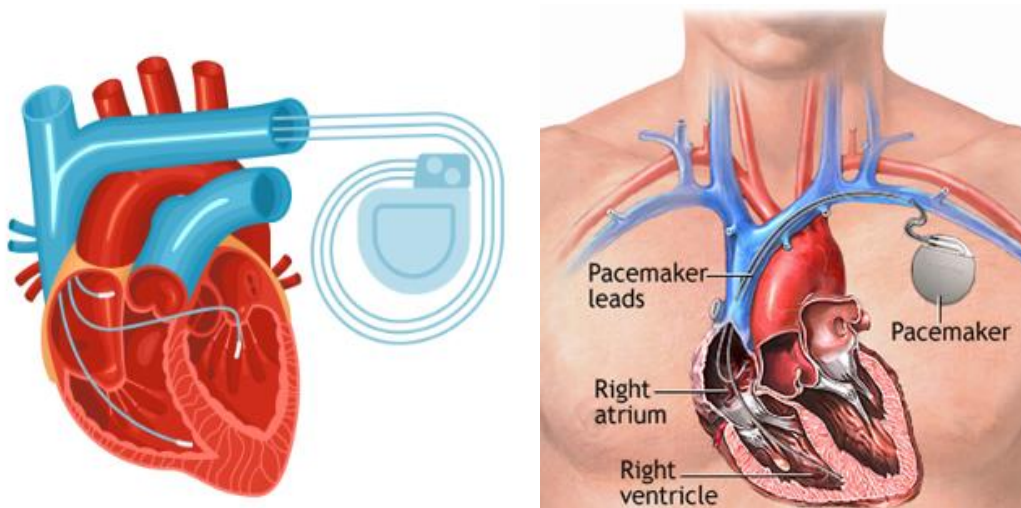


Pacemaker Patient Guide

1. What is a Pacemaker?

A pacemaker is a small medical device implanted under the skin of the chest to help regulate abnormal heart rhythms. It uses electrical impulses to stimulate the heart so it beats at a normal rate and rhythm. Pacemakers are commonly recommended for people with arrhythmias, particularly when the heart beats too slowly (bradycardia).



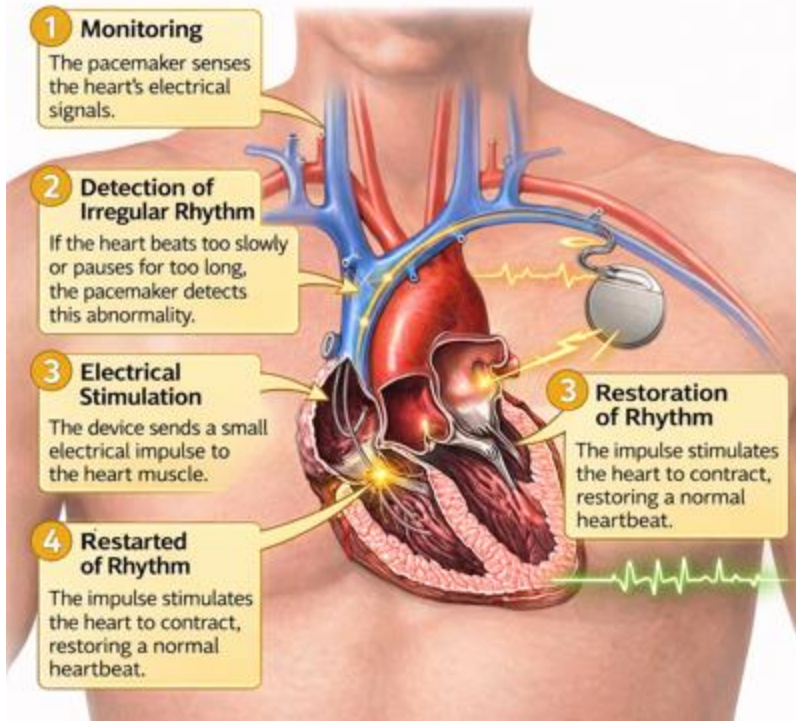
2. How a Pacemaker Works

A pacemaker continuously monitors the heart's electrical activity. If it detects that the heart is beating too slowly or irregularly, it sends a small electrical impulse to stimulate the heart muscle. This impulse restores a normal heartbeat and ensures proper blood circulation.

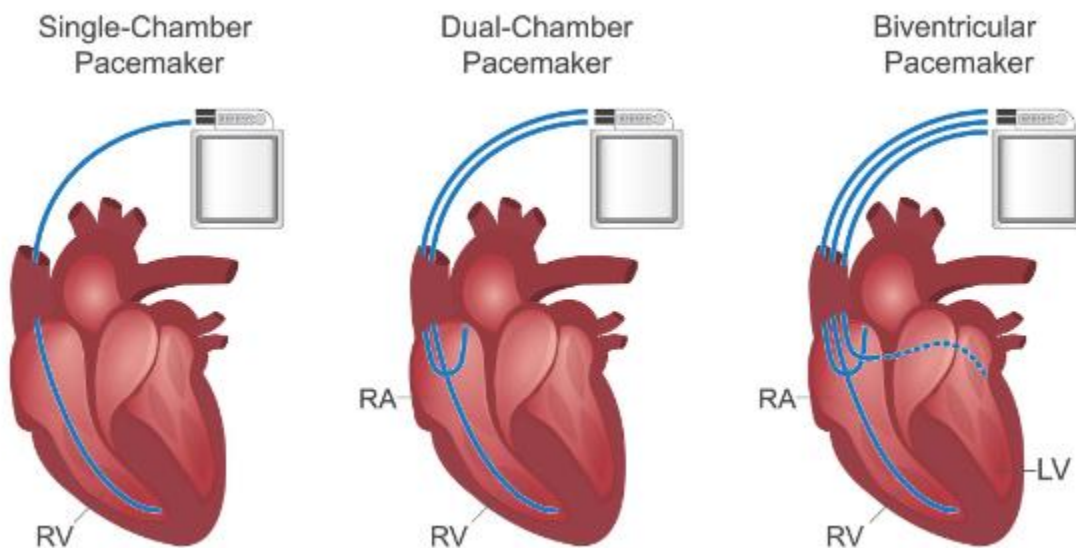
Steps:

1. The device monitors the heart rhythm.
2. If an abnormal rhythm is detected, the pacemaker activates.
3. Electrical signals are delivered through leads to the heart.
4. The heart contracts and normal rhythm is restored.

How a Pacemaker Works



3. Types of Pacemakers



Dual-Chamber Pacemaker

Uses two leads: one in the right atrium and one in the right ventricle to coordinate timing between chambers.

Single-Chamber Pacemaker

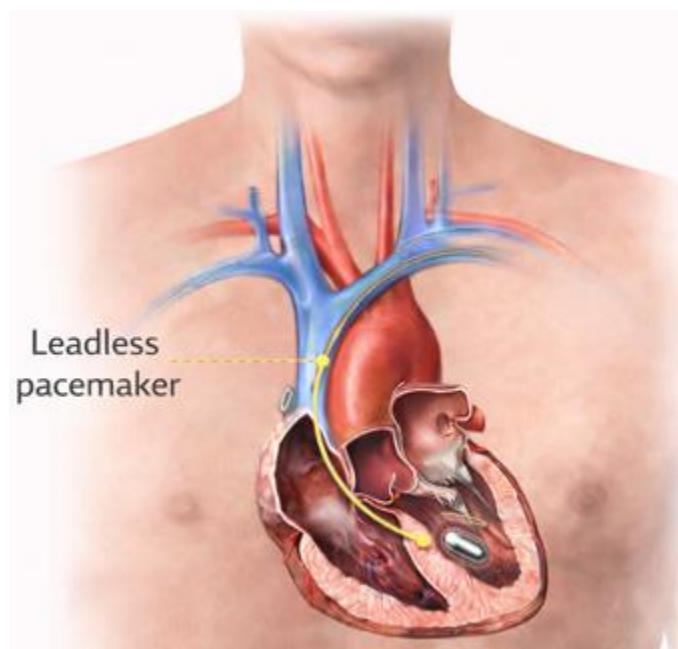
Uses one lead placed in either the right atrium or right ventricle to regulate heartbeat.

Biventricular Pacemaker (CRT)

Also called Cardiac Resynchronization Therapy. This type stimulates both ventricles and is used in some heart failure patients.

Leadless Pacemaker

A newer miniature pacemaker placed directly inside the heart without traditional leads.



4. Pre-Procedure Preparation

Before pacemaker implantation, doctors typically conduct tests and provide preparation instructions.

Common steps include:

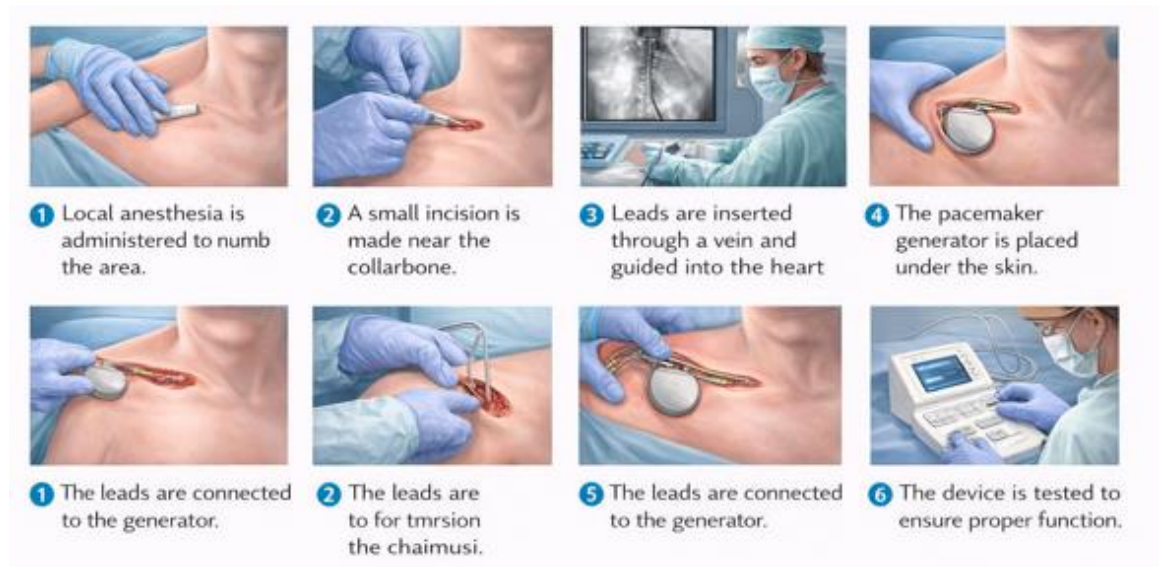
- Blood tests
- Electrocardiogram (ECG)
- Chest X-ray
- Medication review
- Fasting before surgery if instructed
- Arranging transportation home after the procedure

5. What to Expect During the Procedure

The pacemaker implantation procedure usually takes 1–2 hours and is performed in a hospital setting.

Typical steps include:

1. Local anesthesia is administered.
2. A small incision is made near the collarbone.
3. Leads are inserted through a vein into the heart.
4. The pacemaker generator is placed under the skin.
5. The system is tested and the incision is closed.



6. Post-Operative Care

1–2 Weeks After Surgery

- Keep incision site clean and dry
- Avoid raising the arm above shoulder height
- Avoid lifting objects heavier than 10 pounds
- Monitor for signs of infection

3–8 Weeks After Surgery

- Gradually return to light daily activities
- Attend follow-up appointments
- Avoid vigorous arm movement or heavy lifting

Beyond 8 Weeks

- Most patients can return to normal activities
- Exercise can resume with physician approval

- Routine device checkups are required
- Pacemaker batteries typically last 7–15 years

7. When to Call the Doctor

Contact your doctor immediately if you experience:

- Fever or chills
- Redness or swelling at the incision site
- Persistent pain around the pacemaker
- Shortness of breath
- Dizziness or fainting
- Rapid or irregular heartbeat

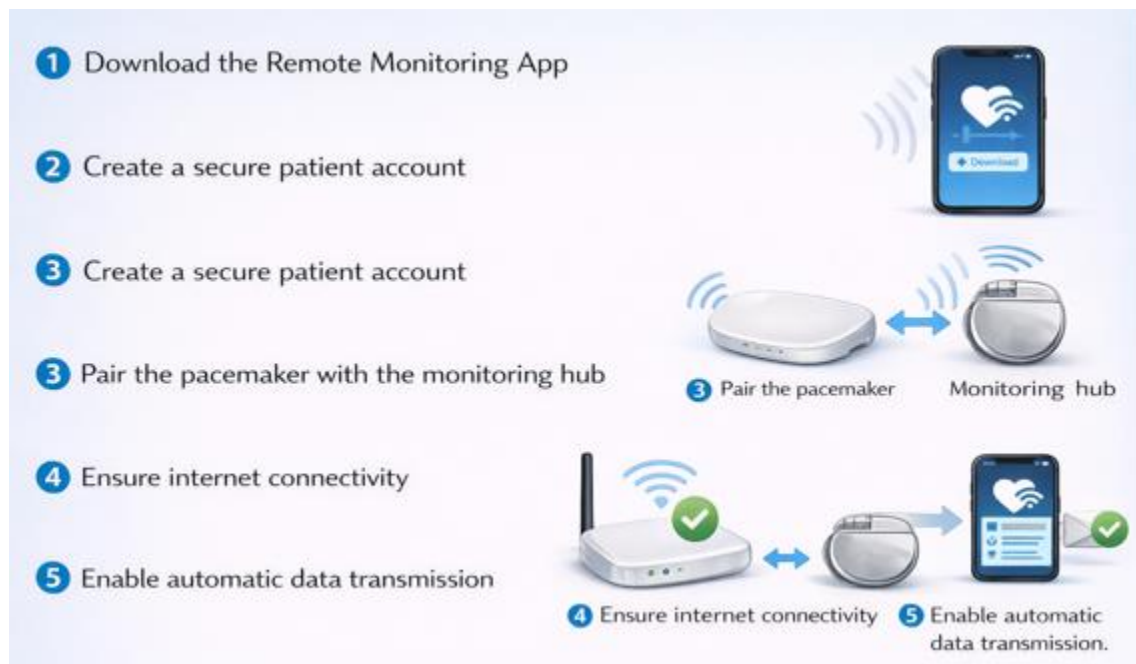
8. Remote Monitoring App

Getting Started

Pacemaker remote monitoring systems allow physicians to monitor device data remotely.

Typical setup steps:

1. Download the **Remote Monitoring App**
2. Create a secure patient account
3. Pair the pacemaker with the monitoring hub
4. Ensure internet connectivity
5. Enable automatic data transmission

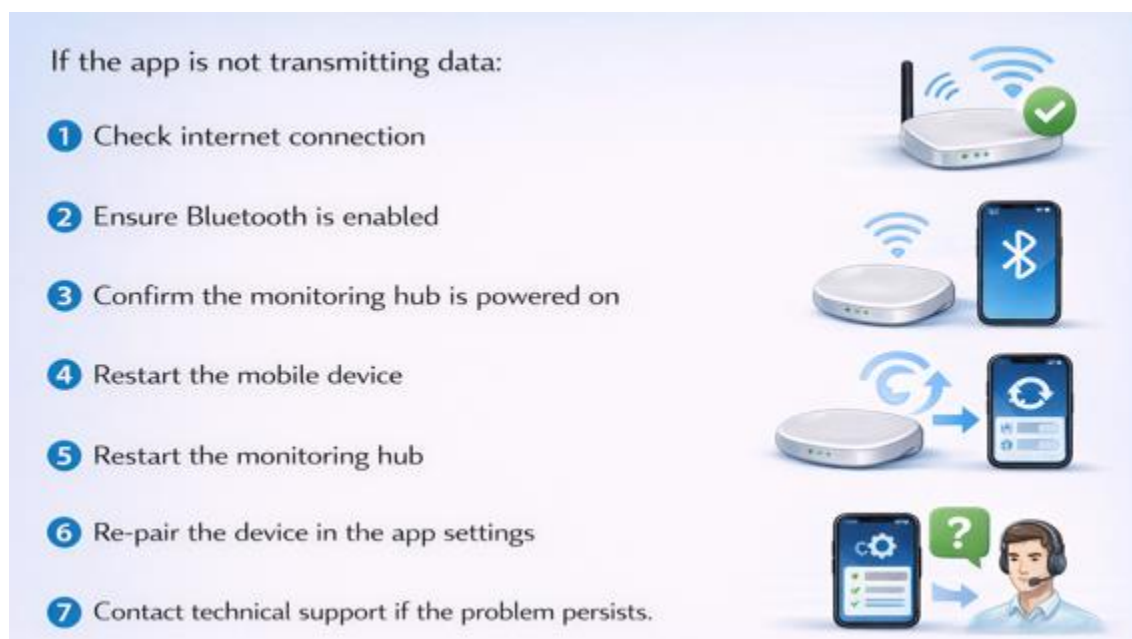


The app sends pacemaker data to healthcare providers for **routine monitoring and alerts**.

Troubleshooting Steps

If the app is not transmitting data:

1. Check internet connection.
2. Ensure Bluetooth is enabled.
3. Confirm the monitoring hub is powered on.
4. Restart the mobile device.
5. Restart the monitoring hub.
6. Re-pair the device in the app settings.
7. Contact technical support if the problem persists.

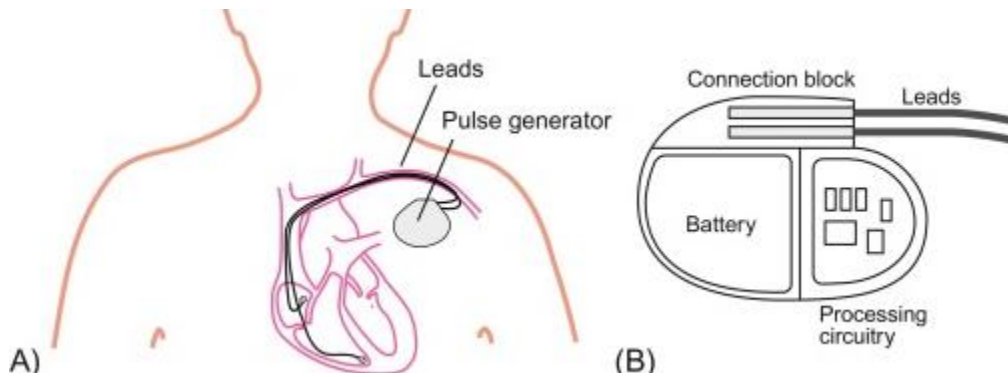


9. Instruction for Use

A. Pacemaker

Device Description and Function

A pacemaker consists of a pulse generator (battery and electronics) and leads that deliver electrical impulses to the heart muscle and Electrodes Located at the tip of leads to sense and stimulate heart tissue. The system continuously monitors heart rhythm and provides pacing support when necessary.



Electromagnetic Interference (EMI) and Safety

Certain electronic devices may interfere with pacemaker operation. Patients should avoid prolonged exposure to strong electromagnetic fields such as industrial welding equipment or MRI scanners unless approved by a physician.

Regulatory Compliance Information

This pacemaker device is cleared or approved by the U.S. Food and Drug Administration (FDA). The regulatory label on the device packaging indicates the approval or clearance status. Included on Product Packaging:

- FDA approval or clearance status: Indicates whether the device is FDA-approved (PMA) or FDA-cleared (510(k)).
- Model Number: Identifies the specific pacemaker model.
- Serial Number: Unique identifier assigned to each manufactured unit.
- Manufacturer information: Name of the manufacturer, Address (city, state/country), Contact information (phone, website, or customer support)

Device Families

Pacemakers are marketed under **device families** or product lines from manufacturers. Each family may contain multiple models with similar technology and indications.

Indications for Use

Indications for use describe the medical conditions for which the pacemaker is intended.

Typical indications include:

1. Bradycardia

Pacemakers are commonly indicated for patients experiencing **symptomatic bradycardia**, where the heart beats too slowly to maintain adequate circulation.

2. Atrioventricular (AV) Block

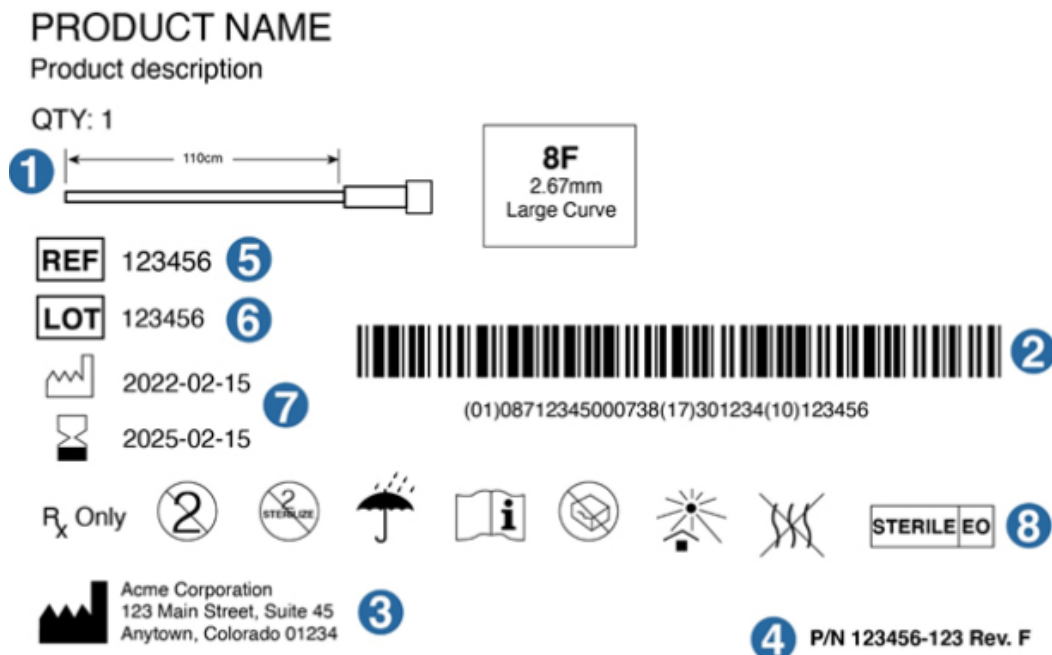
Patients with **second-degree or third-degree AV block**, where electrical signals from the atria do not properly reach the ventricles.

3. Sick Sinus Syndrome

A condition in which the heart's natural pacemaker (sinus node) does not function correctly.

4. Heart Failure with Dyssynchrony

Certain patients with heart failure may benefit from **Cardiac Resynchronization Therapy (CRT)** pacemakers.



Warranty Information

Pacemaker manufacturers typically provide warranty coverage.

Warranty Duration: Pacemaker warranties typically last 5–10 years, depending on the model and manufacturer

1. Warranty May Cover

- Device manufacturing defects
- Battery failure
- Electronic malfunction

2. Warranty usually **does not cover**:

- Damage from improper handling
- Surgical complications unrelated to the device
- Damage due to external trauma

3. Additional Warranty Considerations

- Eligibility Requirements: Device must be implanted, used, and maintained according to manufacturer guidelines.
- Device Return for Analysis: To process a warranty claim, the manufacturer usually requires the returned device for technical evaluation.
- Transferability: Warranty coverage is tied to the device, not the patient, but is applied only for that implant episode.
- Limitations of Manufacturer Liability: Most warranties limit liability to replacement of the product or reimbursement caps; they do not cover all clinical expenses.

Patients should retain their **implant card and warranty documentation**.

Warranty Key Differences:

Aspect	Under Warranty	Out of Warranty
Reason	Defect or malfunction	Normal battery depletion
Device Cost	Usually covered by manufacturer	Covered by insurance/patient
Process	Manufacturer involved	Standard clinical replacement
Urgency	Depends on malfunction severity	Planned (based on battery life)

Pacemaker Warranty – Coverage & Replacement Summary

1.If Replacement Is Needed (Under Warranty)

When This Applies

- Device malfunction
- Early battery failure
- Confirmed manufacturing defect

What to Do

1. Contact your healthcare provider
 - Report symptoms or device alerts
2. Device evaluation
 - Clinic performs device interrogation
 - Confirms issue

3. Warranty verification
 - Provider coordinates with manufacturer
 - Confirms eligibility
4. Replacement procedure
 - New device provided (typically covered)
 - Generator is replaced; leads usually remain

2.If Replacement Is Needed (Out of Warranty)

When This Applies

- Normal battery depletion (end-of-life)
- Device aging or upgrade

What to Do

1. Routine monitoring
 - Device shows ERI (Elective Replacement Indicator)
2. Schedule replacement
 - Physician plans procedure before battery depletion
3. Replacement procedure
 - Generator is replaced
 - Leads reused if functioning properly

B. Remote Monitoring Hub (Home Monitor)

1.What You Need Before Setup

Before setting up the home monitor, ensure you have:

- The **Home Monitor device**
- Power adapter and cable
- Internet connection (Wi-Fi, cellular, or landline depending on model)
- Your **implant card** (device details)
- Instructions provided by your healthcare provider

2. Step-by-Step Setup Instructions

Step 1 — Choose the Right Location

- Place the monitor **near your bed**
- Keep it within **6–10 feet (2–3 meters)** of where you sleep
- Ensure the area is:

- Dry
- Well-ventilated
- Near a power outlet

Step 2 — Connect to Power

- Plug the monitor into a wall outlet
- Turn the device ON (if required)

Check for **power indicator light**

Step 3 — Establish Connectivity

Depending on your device type:

Cellular Model

- Automatically connects (no setup needed)

Wi-Fi Model

- Enter Wi-Fi credentials (if required)

Landline Model

- Connect to telephone line

Step 4 — Pair with Your Pacemaker

- Some systems pair automatically
- Others require:
 - Pressing a **sync/pair button**
 - Following on-screen instructions

Keep the monitor close during pairing.

Step 5 — Perform Initial Transmission

- The monitor may:
 - Automatically send data, OR
 - Prompt you to initiate transmission

Stay near the device until transmission is complete.

Step 6 — Confirm Successful Setup

Check for indicators such as:

- Green light / “Connected” status
- Confirmation message (if screen-enabled)

Your healthcare provider may also confirm receipt of data.

3. Troubleshooting Decision Tree

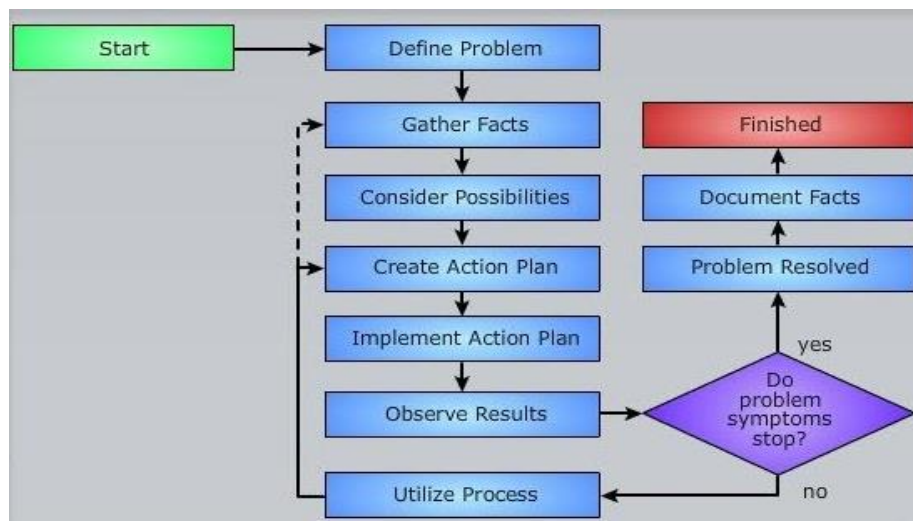
Step 1: Check power supply.

Step 2: Verify internet or cellular connectivity.

Step 3: Confirm the pacemaker is within transmission range.

Step 4: Restart the monitoring hub.

Step 5: Contact technical support if transmission still fails.



4. Daily Usage Guidelines

- Keep the monitor **plugged in at all times**
- Sleep near the device for automatic transmission
- No daily interaction is usually required
- Follow any instructions from your healthcare provider

5. Important Safety Tips

- Do not place the device:
 - Near water
 - In direct sunlight
 - Near strong magnets
- Keep mobile phones at least **6 inches away** from the pacemaker site
- Do not unplug unless instructed

C. Implant Card and Device Label

Patients receive an implant card after surgery.

This card usually includes:

- Device model number
- Serial number
- Manufacturer name
- Implant date
- Physician or hospital information

The same information can be found on the device packaging label.